Swagata Chakraborty
San Francisco Bay Area, CA, USA| (858) 729-3992 | swagatanit@gmail.com | LinkedIn

PROFESSIONAL SUMMARY

Senior Data Scientist (8+ years) specializing in product decisions under ambiguous problem statements & imperfect data. I build experimentation frameworks, causal analyses, & ML/LLM evaluation systems that ensure metric integrity & drive growth across Search, Ads, & marketplaces.

WORK EXPERIENCE

Houzz, Palo Alto, CA - *Data Scientist, Consumer & SEO Lead* **Sep 2022 – Present**

GenAI Search and Conversational Q&A

- Led GenAI analytics for Houzz's AI Q&A experience and increased unique visitor searches by 5% —classifying conversational intent across 5 verticals (inspiration, advice, software, furniture, services) and popular room categories to improve discovery.
- Improved text-image response relevance (+10%) & content freshness (+15%) by partnering with Product, Eng, Editorial teams.

SEM & Paid Marketing Optimization (Revenue & Churn Impact)

- Doubled Return on Ad Spend (RoAS) by redesigning bidding strategy and optimizing lead conversion value (a conversion is a successful hire of a service professional/contractor/remodeler) following an investigation of the YoY SEM performance decline driven by traffic mix shifts, lead quality degradation and changes in keyword trends.
- Reduced spam-driven spend by introducing lead-quality guardrails (spam suppression, outlier capping, dynamic conversion values), targeting a 23% reduction in service contractors churn from Ads subscription by mid 2026.

Message & Lead Sentiment Measurement (LLM-as-Judge)

- Designed and deployed an LLM-as-judge framework to score homeowner–service professional message sentiment, improving positive-label precision by +3% over legacy model.
- Launched the sentiment-adjusted lead count metric for A/B tests and executive KPI dashboards, uncovering quality regressions previously masked by volume-only growth leads.

Service Professionals Trust & Verification: Causal Inferencing

- Led causal-inference experiments evaluating service contractor Merits and License verification, increasing licensed-service contractors' coverage by ~50% in pilot geographies.
- Enabled nationwide rollout and vendor contract renewal by providing statistically defensible lift estimates under real-world marketplace & budget constraints.

Ads Experimentation Platform Infra

- Built a unified metrics pipeline (4x faster queries) and experimentation platform supporting Ads and Marketplace decisions, with multivariate correction, segmentation, and robust outlier handling.
- Enabled sequential testing for noisy, high-variance experiments, accelerating iteration and safe ramp-up; powered launches like the project-type selector that delivered +8% service professional or contractor listing CTR and +2.5% lead funnel lift.

Walmart, Sunnyvale, CA - *Data Scientist, E-commerce* **Oct 2018 – Sept 2022**

E-Commerce Catalog Assortment & Robustness

- Developed production ML system using Siamese network embeddings and hierarchical clustering, improving variant-group recall by 15% across 10 high-revenue categories (Home & Kitchen, Apparel, Sports, etc.)
- Architected bias-reduced training & evaluation datasets; eliminated bi-quarterly manual retraining of the evolving catalog.
- Implemented image-hashing–based anomaly detection script to flag brand & product-type inconsistencies to surface top 1000 offender sellers in revenue-critical categories.

Mazda, Irvine, CA - *Data Analyst Intern, Pricing* **April 2018 – June 2018**

- Built an interactive vehicle wholesale auction-price forecasting tool for ~1M vehicles using XG-Boost and t-SNE clustering.

Citibank, Chennai, India - *Data Analyst, Credit Card Campaigns* **July 2015 – June 2017**

- Streamlined 100+ customer-segmentation campaigns, driving 4% credit-card revenue lift and 8% conversion lift.

PUBLICATIONS

- US Patent: ML-based Product Variant Grouping (Walmart)
- SISAP 2020: Similar Image Retrieval in Walmart Product Search

TECHNICAL SKILLS

Programming: Python, SQL, R, Spark

Data Science: Machine Learning, GenAI (LLM Eval, Prompt Engineering), A/B Tests, Causal Inference, Sequential Testing, Statistics

Data Platforms & BI: Airflow, AWS, GCP, Docker, GitHub, Tableau, Amplitude

EDUCATION

University of California, San Diego, CA - *Master of Sciences, Business Analytics* **Aug 2017 – Sept 2018**

National Institute of Technology (Trichy), India - *Bachelor of Engineering, Electronics & Communication* **July 2011 – April 2015**